Deep Learning Approaches for Automated Diabetic Retinopathy Detection and Classification: A Review of Nine Selected Studies

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***Abstract*—Diabetic retinopathy (DR) is a retinal complication of diabetes in which progressive vascular and lesion-related changes can lead to visual impairment. Automated analysis of retinal fundus photographs has therefore become an important research area, particularly because manual grading can be time-consuming and early-stage abnormalities may be subtle. This paper reviews nine selected studies supplied as the reference corpus. The reviewed works cover convolutional neural networks, transfer learning, EfficientNet architectures, contrast enhancement, hybrid and ensemble learning, multi-label classification, attention mechanisms, and explainable artificial intelligence. The studies illustrate a progression from handcrafted feature extraction and conventional classifiers toward end-to-end deep learning and lightweight attention-based models. Several papers emphasize the importance of preprocessing and transfer learning when datasets are limited or visually heterogeneous, while others address multi-label disease recognition and interpretability. The review also compares the datasets, principal methodologies, reported evaluation measures, and limitations described in the selected studies. Across the corpus, class imbalance, subtle early-stage lesions, image-quality variation, cross-dataset generalization, computational cost, and model transparency remain recurring challenges. The reviewed evidence indicates that effective automated DR analysis benefits from combining robust representation learning with suitable preprocessing, efficient architectures, balanced evaluation, and interpretable outputs.**

***Index Terms*—Diabetic retinopathy, retinal fundus images, deep learning, convolutional neural networks, transfer learning, EfficientNet, CLAHE, attention, explainable AI.**

I. Introduction

Diabetic retinopathy is associated with damage to retinal blood vessels and can progress through multiple severity levels. The reviewed literature commonly describes five stages ranging from no visible retinopathy through mild, moderate, and severe non-proliferative disease to proliferative diabetic retinopathy. The visual manifestations include microaneurysms, hemorrhages, exudates, cotton-wool spots, venous abnormalities, and neovascularization, although their visibility and distinction vary with disease severity and image quality [4], [6], [7], [9].

Retinal fundus photography is widely used for screening because it provides a visual record of retinal structures. Nevertheless, manual interpretation requires trained specialists and may be difficult to scale. The selected literature repeatedly identifies early-stage abnormalities as a particularly challenging problem because lesions can be small, visually similar to other retinal structures, or affected by acquisition conditions [2], [7], [9]. These limitations have motivated computer-aided approaches that learn discriminative representations directly from fundus images.

The nine selected studies represent several stages in the development of automated DR analysis. Conventional image processing and machine-learning combinations are represented by Bilal *et al.* [6], while CNN-based and transfer-learning approaches appear in the works of AbdelMaksoud *et al.* [7], Momeni Pour *et al.* [4], Khan *et al.* [9], and Wang *et al.* [1]. Kim *et al.* provide a broader review of transfer learning for medical image classification [5], and Atwany *et al.* survey deep-learning approaches specifically for DR [2]. More recent directions are represented by Shahzad *et al.*, who investigate explainable AI [3], and Dharrao *et al.*, who combine EfficientNetB0, multi-scale spatial attention, and Grad-CAM [8].

This paper summarizes only these nine supplied studies. It does not introduce experimental results from any external study and does not present a new diagnostic model. The objective is to organize the findings of the selected literature and identify common methodological trends and research challenges.

II. Scope of the Reviewed Literature

The reference corpus contains nine papers published or released between 2020 and 2025. Five are IEEE Access papers, one is a BMC Medical Imaging review, one is a Discover Applied Sciences research article, and one is an arXiv preprint. Their topics range from specific architectures to broad surveys. Table I gives a compact overview.

III. Review of Individual Studies

A. Transfer Learning and Customized EfficientNet

AbdelMaksoud, Barakat, and Elmogy proposed a deep-learning computer-aided system based on a customized EfficientNet model and transfer learning [7]. The study used the Indian Diabetic Retinopathy Image Dataset (IDRiD), which contains multiple retinal lesion labels and severity categories. The authors motivated transfer learning by the limited size of

TABLE I
OVERVIEW OF THE NINE SELECTED STUDIES

Ref.	Main focus	Principal approach	Main reported result / emphasis
[1]	Wang *et al.*	EfficientNet-based multi-label ensemble for fundus diseases	Multi-label recognition using fused model probabilities
[2]	Atwany *et al.*	Survey of supervised, self-supervised and Vision Transformer DR methods	Taxonomy of methods, datasets, metrics, and open challenges
[3]	Shahzad *et al.*	CNN combined with explainable-AI methods for transparent DR analysis	Reported 94% diagnostic accuracy with emphasis on interpretability
[4]	Momeni Pour *et al.*	CLAHE preprocessing with EfficientNet-B5; cross-dataset evaluation	AUC 0.945 on Messidor and 0.932 on IDRiD in the reported referral settings
[5]	Kim *et al.*	Review of transfer learning for medical image classification	121 eligible studies; evidence supporting transfer learning under data scarcity
[6]	Bilal *et al.*	Preprocessing, lesion features, SVM/KNN/BT and majority voting	98.06% accuracy; 83.67% sensitivity; 100% specificity
[7]	AbdelMaksoud *et al.*	Customized EfficientNet with transfer learning; multi-label DR grading on IDRiD	86% accuracy; 78.45% training DSC
[8]	Dharrao *et al.*	EfficientNetB0 + MSAG + Grad-CAM with preprocessing and augmentation	QWK 0.923, macro-AUC 0.976, accuracy 88.4% on APTOS 2019
[9]	Khan *et al.*	VGG16 + Spatial Pyramid Pooling + Network-in-Network (VGG-NiN)	Lower learnable-parameter count; micro-AUC 0.95

medical image datasets and sought to avoid extensive handcrafted feature extraction. The preprocessing pipeline included filtering, resizing to 256×256, cropping, rotation, mirroring, and normalization. A customized EfficientNetB0 configuration was then used for multi-label classification. The study reports 86% accuracy and a Dice similarity coefficient of 78.45%. Experiments with different optimizers, learning rates, and epoch counts were also described, with Adamax at a learning rate of 0.002 and 100 epochs producing the highest reported training DSC. The paper concludes that transfer learning and augmentation are particularly useful for small datasets, while noting the possibility of extending the approach to additional retinal diseases [7].

The study is significant because it illustrates how a pretrained network can reduce the dependence on large task-specific datasets. It also demonstrates that multi-label formulation can represent the simultaneous presence of retinal abnormalities. However, the work remains dependent on a relatively small dataset and reports performance primarily through accuracy and DSC, limiting direct comparison with studies that use other metrics.

B. CLAHE and EfficientNet-B5

Momeni Pour *et al.* investigated automatic DR monitoring using contrast-limited adaptive histogram equalization (CLAHE) followed by EfficientNet-B5 classification [4]. The authors emphasized that retinal images can contain uneven illumination and low-contrast pathological information. CLAHE was used to improve local contrast while limiting excessive noise amplification. EfficientNet was selected because its compound scaling strategy jointly considers network depth, width, and image resolution.

A notable aspect of the study is its cross-dataset evaluation. A model trained using a mixture of Messidor-2 and IDRiD was evaluated on Messidor, while another configuration trained using Messidor-2 and Messidor was evaluated on IDRiD. In the reported referral-oriented experiments, the model achieved an AUC of 0.945 on Messidor with 92% sensitivity and an AUC of 0.932 on IDRiD with 93% sensitivity [4]. The study therefore highlights both preprocessing and dataset diversity as important factors for generalization.

C. VGG-NiN Architecture with Spatial Pyramid Pooling

Khan *et al.* proposed VGG-NiN, which combines VGG16, a Spatial Pyramid Pooling (SPP) layer, and Network-in-Network (NiN) modules for five-stage DR classification [9]. The motivation was to retain useful information from fundus images while reducing the limitations imposed by fixed-size fully connected layers. SPP converts feature maps of varying spatial dimensions into a fixed-size representation, thereby reducing the need for aggressive cropping or resolution reduction. NiN modules are used to provide additional nonlinear feature transformations.

The study used labeled EyePACS/Kaggle fundus images and addressed the class imbalance problem through its experimental evaluation. Transfer learning was applied by freezing the trained VGG convolutional layers while fine-tuning the fully connected portion, with the NiN component initialized separately. The authors reported comparable or improved performance with substantially fewer learnable parameters than a referenced ensemble approach. The micro-average AUC was reported as 0.95, while the authors emphasized that their architecture used 52% fewer parameters while obtaining a similar micro-AUC to the comparison model [9]. A recurring limitation identified in the literature surrounding this work is the difficulty of accurately distinguishing early stages, particularly mild DR.

D. Hybrid Feature Extraction and Majority Voting

Bilal *et al.* presented a mixed-model approach in which image preprocessing and lesion-oriented feature extraction were followed by SVM, KNN, and binary-tree classifiers [6]. The preprocessing stage converted RGB images to YCbCr,

operated on the luminance component, applied median filtering, contrast stretching, and intensity normalization, and reconstructed the RGB representation. Separate processing paths were then used for red and bright lesions. Region statistics such as region count, mean area, mean perimeter, and mean solidity were extracted and fused before classification.

The three classifiers independently produced predictions and a majority-voting mechanism generated the final classification. Across the disease grading database used by the authors, the reported accuracy was 98.06%, sensitivity was 83.67%, and specificity was 100% [6]. The paper demonstrates the value of combining complementary classifiers and lesion-specific features. At the same time, the pipeline requires explicit feature engineering and multiple image-processing stages, which can increase system complexity compared with end-to-end deep-learning approaches.

E. Transfer Learning in Medical Image Classification

Kim *et al.* conducted a literature review focused on transfer learning for medical image classification rather than DR alone [5]. The authors retrieved 425 peer-reviewed articles from PubMed and Web of Science and identified 121 studies that met their inclusion criteria. The review considered feature-extractor approaches, hybrid feature extraction, fine-tuning, and training from scratch, together with different backbone architectures.

The review found that many studies empirically compared multiple backbone models and transfer-learning strategies. In the reported corpus, Inception was the most frequently employed deep model, while feature extraction and fine-tuning-related strategies were common. The authors concluded that transfer learning is effective in the presence of limited medical data and can reduce computational cost and training time. They particularly encouraged the use of deep pretrained models as feature extractors when predictive performance can be retained while reducing resource requirements [5]. For retinal imaging, this provides a broader methodological basis for selecting pretrained CNNs.

F. EfficientNet for Multi-Label Fundus Disease Recognition

Wang *et al.* addressed the more general problem of recognizing multiple fundus diseases in a single image using an EfficientNet-based ensemble [1]. Each constituent model contains an EfficientNet feature extractor followed by a custom classification network designed for multi-label prediction. Probabilities from multiple models are fused to obtain the final result. The study used the ODIR 2019 dataset and emphasized that multi-label classification is more representative of situations in which more than one ocular condition can appear in the same fundus image.

The work demonstrates the flexibility of EfficientNet as a feature extraction backbone beyond single-disease classification. It also emphasizes that an ensemble can combine complementary predictions from several models. However, ensemble designs increase computational and implementation complexity. The paper's primary contribution is therefore methodological: it shows how transfer learning, EfficientNet representations, and probability fusion can be combined for multi-label fundus analysis [1].

G. Survey of Deep Learning Techniques for DR

Atwany *et al.* provided a broad survey of deep-learning techniques for DR classification and detection [2]. The review considers supervised learning, self-supervised learning, and Vision Transformer approaches, together with retinal datasets and different classification targets such as referable, non-referable, and proliferative disease. It discusses the evolution from conventional feature engineering toward deep representation learning and emphasizes the increasing role of attention and transformer-based methods.

The survey identifies several persistent difficulties. Early lesions are subtle; datasets are often imbalanced; image quality and field-of-view differences affect performance; and models may fail to generalize between populations and acquisition conditions. The authors also discuss the importance of evaluation datasets and the need to investigate newer learning paradigms. As a survey, the paper does not propose one single classifier, but it provides a useful map of the methodological landscape and highlights areas where additional validation and research are needed [2].

H. Explainable AI for Transparent DR Analysis

Shahzad *et al.* focused on the interpretability problem associated with deep-learning models for DR [3]. The paper notes that high-performing AI systems can behave as black boxes, making it difficult for healthcare providers to understand why a particular decision was produced. The proposed approach combines deep learning with explainable-AI techniques to provide more comprehensible information about model decisions.

The study reports a diagnostic accuracy of 94% and presents convergence behavior to support the learning process. Its main emphasis, however, is not simply numerical performance but transparency and applicability in healthcare. The authors argue that explanations can help users inspect model reasoning and support confidence in AI-assisted decisions. The paper therefore broadens DR research beyond classification accuracy by treating interpretability as a practical requirement for healthcare deployment [3].

I. EfficientNetB0 with Multi-Scale Spatial Attention and Grad-CAM

Dharrao *et al.* presented a lightweight EfficientNetB0-based framework incorporating Multi-Scale Spatial Attention Gates (MSAG) and Grad-CAM visualization [8]. The study used APTOS 2019 and applied preprocessing involving green-channel emphasis, CLAHE, normalization, resizing, and noise reduction. Augmentation included rotation, flipping, translation, and photometric changes, while class balancing was considered during training.

MSAG was inserted after the final convolutional block to combine spatial information at multiple scales. The attention mechanism computes spatial weights and reweights feature

maps, with the stated goal of emphasizing pathological regions such as microaneurysms and exudates. Grad-CAM was then used to generate visual evidence of influential image regions. On APTOS 2019, the paper reports a mean QWK of 0.923 $\pm$ 0.008, a held-out test QWK of 0.919, macro-AUC of 0.976, and overall accuracy of 88.4% [8]. The model was also described as lightweight, with approximately 5.3 million parameters.

The study explicitly recognizes limitations related to class imbalance, limited dataset scope, cross-dataset generalization, and the need for expert validation of visual explanations. It therefore represents a combination of efficiency, attention, and explainability within a single DR classification framework.

IV. Comparative Methodological Analysis

A. Evolution of Learning Strategies

The nine papers show a clear movement from manually designed image-processing and feature-engineering pipelines toward learned feature representations. Bilal *et al.* rely on lesion detection, region statistics, and conventional classifiers [6]. The transfer-learning studies use pretrained CNN representations to reduce data and computational requirements [4], [7], [9]. More recent work extends this paradigm with attention and explainability [3], [8].

This progression does not imply that conventional methods are universally inferior. The hybrid method of Bilal *et al.* reports strong numerical performance, while explicit lesion features can make parts of the pipeline easier to inspect. However, end-to-end deep models reduce the need to manually define which visual characteristics should be extracted, which is attractive for complex retinal images.

B. Role of Preprocessing

Preprocessing is a recurring component of the selected literature. CLAHE is explicitly central to Momeni Pour *et al.* [4] and also appears in the more recent EfficientNetB0/MSAG framework [8]. Other studies use resizing, normalization, filtering, cropping, flipping, or rotation [6], [7], [9]. The motivation is consistent: fundus photographs may differ in illumination, contrast, noise, field of view, and acquisition conditions.

The literature also shows that preprocessing must be selected carefully. Momeni Pour *et al.* explain that CLAHE can improve local contrast but inappropriate parameters may amplify undesirable effects [4]. This suggests that preprocessing is not merely a cosmetic step; it can alter the visual evidence available to a classifier.

C. Transfer Learning and Model Efficiency

Transfer learning is one of the strongest common themes. AbdelMaksoud *et al.* use a customized EfficientNet for IDRiD [7], Momeni Pour *et al.* use EfficientNet-B5 [4], and Khan *et al.* use pretrained VGG16 components [9]. Kim *et al.*'s broader review provides supporting evidence that pretrained networks are valuable in medical image analysis where labeled data can be limited [5].

EfficientNet appears repeatedly because its architecture is designed around an accuracy-efficiency trade-off. The reviewed papers use different members or configurations of the family, demonstrating that the same general architectural principle can support different dataset sizes and computational settings [1], [4], [7], [8].

D. Handling Class Imbalance and Difficult Stages

Class imbalance is repeatedly identified as a problem, especially for multi-class DR grading. Khan *et al.* explicitly analyze imbalanced data and emphasize the difficulty of distinguishing early stages [9]. Dharrao *et al.* similarly report differences in class-wise performance and identify imbalance and inter-class similarity as sources of errors [8]. Atwany *et al.* discuss class imbalance as a broader challenge across DR research [2].

The literature uses different responses, including augmentation, class-balanced sampling, ensemble methods, and model design. No single strategy is established by these nine papers as universally sufficient. Instead, the evidence suggests that class distribution should be considered during preprocessing, training, and evaluation.

E. Interpretability

Interpretability becomes increasingly prominent in the more recent studies. Shahzad *et al.* explicitly address the black-box nature of deep models through explainable AI [3]. Dharrao *et al.* combine attention with Grad-CAM to visualize influential retinal regions [8]. Khan *et al.* also discuss localization-related ideas through the SPP/NiN architecture and comparison with previous work on lesion-focused analysis [9].

The reviewed evidence suggests that visual explanations can complement numerical metrics, but explanations should not automatically be interpreted as proof of clinical correctness. Dharrao *et al.* specifically identify the need for expert validation and fairness evaluation of attention maps [8].

V. Datasets and Evaluation Practices

The selected studies use several important retinal datasets, including IDRiD, Messidor, Messidor-2, APTOS 2019, EyePACS/Kaggle, and ODIR 2019 [1], [4], [7]–[9]. Dataset differences are important because image quality, acquisition devices, population characteristics, labeling protocols, and class distributions can vary substantially.

The papers also employ different evaluation measures. Accuracy is reported in several studies, while AUC and sensitivity are emphasized in Momeni Pour *et al.* [4]. Khan *et al.* report accuracy, AUC, ROC-related analysis, precision, specificity, and F1-score [9]. AbdelMaksoud *et al.* use accuracy and DSC [7], while Dharrao *et al.* use QWK, AUC, accuracy, F1-score, and confusion matrices [8]. Bilal *et al.* report accuracy, sensitivity, and specificity [6].

These differences mean that the numerical results should not be interpreted as a direct ranking of the nine systems. A reported accuracy from one dataset and task is not directly equivalent to an AUC or QWK from another dataset and

TABLE II
METHODOLOGICAL COMPARISON

Study	Main methodological characteristic	Key challenge addressed
Wang *et al.* [1]	EfficientNet ensemble for multi-label recognition	Multiple fundus diseases
Atwany *et al.* [2]	Survey of DL, self-supervision and transformers	Research gaps and datasets
Shahzad *et al.* [3]	Deep learning + XAI	Model transparency
Momeni Pour *et al.* [4]	CLAHE + EfficientNet-B5 with cross-dataset testing	Contrast and generalization
Kim *et al.* [5]	Systematic transfer-learning review	Data scarcity and TL selection
Bilal *et al.* [6]	Lesion features + SVM/KNN/BT voting	Feature characterization
AbdelMaksoud *et al.* [7]	Transfer learning with customized EfficientNet and augmentation	Limited data, multi-label grading
Dharrao *et al.* [8]	EfficientNetB0 + MSAG + Grad-CAM	Efficiency, localization, explainability
Khan *et al.* [9]	VGG16 + SPP + NiN	Fixed input and parameter efficiency

task. Cross-dataset validation, consistent class definitions, and multiple class-sensitive metrics are therefore important when comparing future systems.

VI. RESEARCH CHALLENGES IDENTIFIED ACROSS THE NINE PAPERS

The reviewed papers collectively identify several unresolved issues.

Early-stage discrimination: Mild DR can contain subtle microaneurysms that are difficult to distinguish from normal retinal structures. This problem is explicitly discussed in the studies by AbdelMaksoud *et al.*, Khan *et al.*, and Atwany *et al.* [2], [7], [9].

Class imbalance: Severe and proliferative categories can contain fewer examples than normal or moderate categories. Imbalance can therefore make aggregate accuracy misleading and can increase errors in minority classes [2], [8], [9].

Image variability: Differences in illumination, camera characteristics, field of view, noise, and image quality affect retinal appearance. CLAHE, normalization, filtering, and augmentation are repeatedly used to reduce these effects [4], [7], [8].

Cross-dataset generalization: Results obtained from one dataset may not transfer directly to another. Momeni Pour *et al.* explicitly use cross-dataset evaluation, while Dharrao *et al.* identify broader external validation as an important limitation [4], [8].

Computational efficiency: Complex ensembles can improve predictive behavior but require additional computation. Khan *et al.* emphasize learnable-parameter reduction, while Dharrao *et al.* target a lightweight EfficientNetB0 architecture [8], [9].

Interpretability and trust: Numerical predictions alone do not reveal the visual evidence used by a model. Shahzad *et al.* and Dharrao *et al.* show the growing importance of explainability, while also indicating that explanation quality needs careful assessment [3], [8].

VII. DISCUSSION

The nine studies demonstrate that automated DR analysis is not dependent on a single modeling strategy. Instead, the literature combines complementary ideas. Traditional image processing can improve relevant structures, transfer learning can reduce dependence on limited labeled data, efficient CNN architectures can reduce computational cost, ensembles can combine predictions, and explainability techniques can provide information about model behavior.

A particularly visible trend is the development of efficient architectures. EfficientNet appears in multiple studies, including customized classification on IDRiD [7], EfficientNet-B5 with CLAHE and cross-dataset evaluation [4], multi-label fundus disease recognition [1], and EfficientNetB0 with MSAG and Grad-CAM [8]. This shows how a common backbone family can be adapted to different objectives and computational settings.

Another important trend is the movement from isolated performance reporting toward broader evaluation. The reviewed papers use accuracy, AUC, sensitivity, specificity, F1-score, DSC, QWK, ROC curves, precision-recall curves, and confusion matrices. This is valuable because DR grading is a multi-class and imbalanced problem. Multiple complementary metrics therefore provide a more informative description than one aggregate score. Sensitivity and specificity are particularly relevant for screening-oriented evaluation, while class-sensitive measures such as F1-score, DSC, and QWK provide additional information for severity grading.

The selected papers also indicate that model accuracy alone is insufficient for practical healthcare adoption. Interpretability, computational requirements, robustness to image variability, and external validation are recurring concerns. The work incorporating attention and Grad-CAM connects predictive performance with visual evidence, while the XAI-focused study makes transparency a central objective [3], [8]. Visual explanations should nevertheless be treated as supporting evidence rather than independent proof of clinical correctness.

Different methodological choices address different parts of the overall problem. Preprocessing addresses variations in image quality, transfer learning addresses limited annotated data, efficient architectures address computational constraints, and attention or explainability methods address the need to identify relevant image regions. Thus, future systems may benefit from combining complementary components rather than optimizing a single stage in isolation.

The comparison also suggests that future evaluations should report sufficient context about datasets, class distributions, preprocessing strategies, and evaluation protocols. Such reporting would help distinguish methodological improvements from differences caused by experimental settings and would make it easier to assess minority-class behavior and cross-dataset robustness.

The datasets used across the selected studies differ in size, disease distribution, acquisition conditions, and labeling characteristics. Consequently, strong performance on a single dataset cannot by itself establish broad robustness. The cross-dataset experiments of Momeni Pour *et al.* provide an important example of evaluating performance beyond the training distribution [4], while the broader surveys reinforce the need for consistent validation [2], [5].

The literature further indicates that efficiency and interpretability are becoming increasingly important alongside predictive performance. A model with high accuracy but substantial computational requirements may be difficult to deploy in resource-constrained settings, while a prediction without useful supporting evidence may be difficult for users to assess. EfficientNet, attention, and explainability-oriented studies demonstrate approaches for addressing these concerns [1], [3], [8].

Taken together, these observations indicate that automated DR systems should be evaluated as complete pipelines rather than isolated classifiers. Image quality handling, class distribution, learned representation, classification performance, computational requirements, and interpretability all contribute to practical usefulness. This integrated perspective provides a clearer basis for comparing the approaches represented by the nine selected studies.

VIII. Conclusion

Nine selected studies cover automated diabetic retinopathy detection and classification. The literature spans conventional lesion-based feature engineering, CNNs, transfer learning, EfficientNet architectures, multi-label and ensemble methods, attention mechanisms, and explainable AI. Across the corpus, subtle early-stage lesions, class imbalance, image-quality variation, limited labeled data, computational constraints, cross-dataset generalization, and interpretability remain important challenges.

The literature shows that preprocessing and transfer learning can provide useful foundations for retinal image analysis, while EfficientNet-based models offer an efficiency-oriented direction. Hybrid and ensemble approaches demonstrate the value of combining complementary representations or classifiers, whereas attention and explainability methods address the need to understand model behavior. However, numerical results from different studies should be interpreted in the context of their datasets, tasks, class definitions, and evaluation metrics rather than as a direct leaderboard.

Overall, the nine-paper corpus indicates a shift toward systems that aim to be accurate, efficient, robust, and interpretable. Future research should emphasize standardized evaluation, external validation across populations and imaging conditions, minority-class performance, and meaningful validation of explanation methods. Progress therefore depends not only on selecting a stronger network, but also on appropriate preprocessing, representative datasets, balanced evaluation, and careful validation.

An additional implication is the importance of distinguishing experimental performance from practical applicability. Differences in preprocessing, class distribution, and validation strategy can affect measured outcomes substantially. Future studies should therefore report sufficient experimental context to support fair comparison and reproducibility. Class-wise performance and confusion patterns should also be reported where possible, since aggregate accuracy may conceal errors concentrated in clinically important severity categories.

Finally, the increasing use of attention and explainable-AI methods suggests that future automated DR analysis will consider not only the prediction produced but also the evidence associated with that prediction. Such explanations require careful validation and should not be treated as direct evidence of clinical correctness. Combining predictive evaluation with appropriate validation of explanations can support more transparent and reliable computer-aided retinal analysis systems.

References

[1] J. Wang, L. Yang, Z. Huo, W. He, and J. Luo, "Multi-Label Classification of Fundus Images With EfficientNet," *IEEE Access*, vol. 8, pp. 212499–212510, 2020, doi: 10.1109/ACCESS.2020.3040275.

[2] M. Z. Atwany, A. H. Sahyoun, and M. Yaqub, "Deep Learning Techniques for Diabetic Retinopathy Classification: A Survey," *IEEE Access*, vol. 10, pp. 28642–28655, 2022, doi: 10.1109/ACCESS.2022.3157632.

[3] T. Shahzad, M. Saleem, M. S. Farooq, S. Abbas, M. A. Khan, and K. Ouahada, "Developing a Transparent Diagnosis Model for Diabetic Retinopathy Using Explainable AI," *IEEE Access*, vol. 12, 2024, doi: 10.1109/ACCESS.2024.3475550.

[4] A. Momeni Pour, H. Seyedarabi, S. H. Abbasi Jahromi, and A. Javadzadeh, "Automatic Detection and Monitoring of Diabetic Retinopathy Using Efficient Convolutional Neural Networks and Contrast Limited Adaptive Histogram Equalization," *IEEE Access*, vol. 8, pp. 136668–136673, 2020, doi: 10.1109/ACCESS.2020.3005044.

[5] H. E. Kim, A. Cosa-Linan, N. Santhanam, M. Jannesari, M. E. Maros, and T. Ganslandt, "Transfer learning for medical image classification: a literature review," *BMC Medical Imaging*, vol. 22, art. no. 69, 2022, doi: 10.1186/s12880-022-00793-7.

[6] A. Bilal, G. Sun, Y. Li, S. Mazhar, and A. Q. Khan, "Diabetic Retinopathy Detection and Classification Using Mixed Models for a Disease Grading Database," *IEEE Access*, vol. 9, pp. 23544–23553, 2021, doi: 10.1109/ACCESS.2021.3056186.

[7] E. AbdelMaksoud, S. Barakat, and M. Elmogy, "Diabetic Retinopathy Grading System Based on Transfer Learning," arXiv:2012.12515, 2020.

[8] D. Dharrao, M. Dharrao, S. Patil, S. Salvin, P. Ahire, and Y. Dongre, "AI-driven detection and classification of diabetic retinopathy stages using EfficientNetB0," *Discover Applied Sciences*, vol. 7, art. no. 1400, 2025, doi: 10.1007/s42452-025-07998-9.

[9] Z. Khan, F. G. Khan, A. Khan, Z. U. Rehman, S. Shah, S. Qummar, F. Ali, and S. Pack, "Diabetic Retinopathy Detection Using VGG-NIN Deep Learning Architecture," *IEEE Access*, vol. 9, pp. 61408–61420, 2021, doi: 10.1109/ACCESS.2021.3074422.